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MQA-NET: MULTI-QUERY ADAPTIVE NETWORK FOR GENERALIZABLE MEDICAL IMAGE SEGMENTATION

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ABSTRACT

Medical image segmentation has made significant progress thanks to the rapid advancements in deep learning technologies. However, when the model is applied to cross-center medical images, the domain shift often leads to a significant drop in performance. Additionally, common challenges in medical imaging, such as ambiguous boundaries and heterogeneous or visually similar background regions, become more pronounced due to domain shift, further contributing to performance degradation. These challenges hinder the deployment of deep neural networks in real-world clinical settings. To overcome them, we propose the Multi-Query Adaptive Network (MQA-Net) for generalizable medical image segmentation. MQA-Net features an Adaptive Prompt Query (APQ) module that generates input-specific prompts, enhancing the model's adaptability to varying data. Additionally, its Multi-Query Enhanced Transformer Decoder (MQETD) module, in addition to the semantic query, incorporates boundary-oriented and true-false-positive queries for each semantic category, enhancing target boundary segmentation precision and suppressing false-positive regions. Extensive experiments on two medical segmentation tasks demonstrate that MQA-Net significantly outperforms state-of-the-art methods, showing superior generalization capabilities in unseen domains.

Index Terms— Medical image segmentation, Domain generalization, Query-based Transformer

1. INTRODUCTION

Medical image segmentation [1–3] plays a crucial role in clinical settings by assisting physicians and improving operational efficiency. With advances in deep learning, significant progress has been made in automatic medical image segmentation. Notably, convolutional neural networks (CNNs) such as U-Net [4] and Vision Transformers (ViT) [5] have achieved remarkable success. However, they face a critical challenge: domain shift. In practice, data distributions vary substantially across medical centers due to differences in scanners, imaging protocols, and patient populations. This discrepancy leads to severe performance degradation when a model trained in one domain is applied to an unseen domain.

Domain Generalization (DG) [6–10] aims to mitigate performance degradation from domain shifts, enabling models to generalize to unseen target domains. Existing methods are typically grouped into three categories: data-level methods [11–13] that enrich training-data diversity via augmentation or generative style transfer; feature-level methods [14, 15] that learn domain-invariant representations by aligning feature distributions or incorporating domain-agnostic priors (e.g., shape); and model-level methods [16–20] that enhance adaptability through meta-learning, adversarial training, and related strategies. However, most approaches focus on the input/feature spaces, striving for domain-invariant representations, while largely overlooking domain-specific biases and shifts that manifest in the output space.

We propose a Multi-Query Adaptive Network(MQA-Net) for generalized medical image segmentation, designed to address common challenges in cross-center tasks: significant distribution shifts, blurred object boundaries, and intensity similarities between targets and the background. Specifically, the network introduces an adaptive prompt query module to better handle diverse data distributions. Furthermore, it features a multi-query enhanced Transformer decoder that, in addition to the original semantic query, incorporates two specialized queries to respectively tackle blurred boundaries and minimize false-positive predictions. To ensure accurate association and differentiation among these queries, we employ a masked self-attention mechanism. The contributions of this method are as follows:

- We propose a novel MQA-Net to address key challenges in cross-center medical data, including substantial domain shifts, heterogeneous background, and ambiguous target boundaries.

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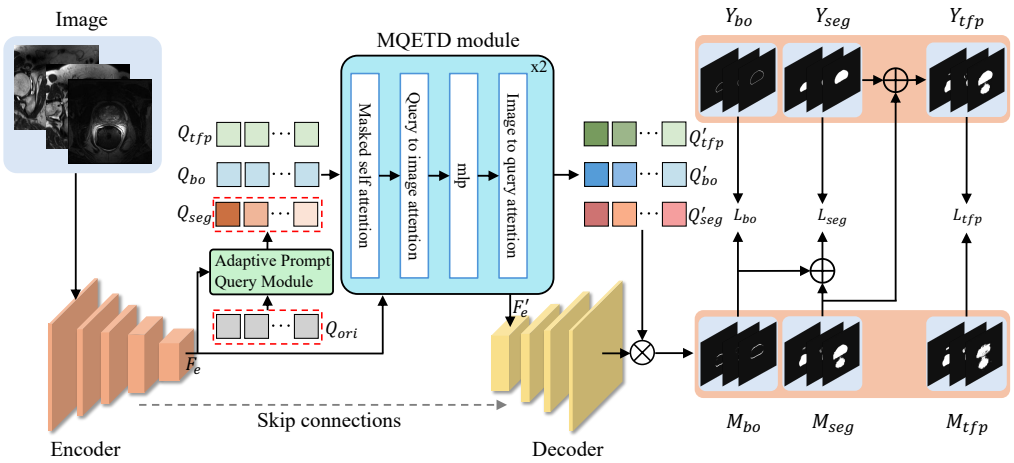


Fig. 1: Illustrating of our MQA-Net: Comprising a U-Net-like architecture, an Adaptive Prompt Query Module (APQ), and a Multi-Query Enhanced Transformer Decoder(MQETD) Module.

- We introduce an APQ module that enhances the model's robustness to significant cross-domain data variations, enabling it to dynamically adapt to each sample.
- We design a multi-query enhanced Transformer decoder module with a masked self-attention layer, enabling the model to segment anatomically accurate targets.

2. METHODOLOGY

The architecture of MQA-Net, illustrated in Fig. 1, primarily consists of three components: 1) a U-Net-like basic structure, 2) an adaptive prompt query(APQ) module, and 3) a multi-query enhanced Transformer decoder(MQETD) module. This U-Net-like architecture comprises an encoder and a decoder, interconnected by skip connections. The details of these modules will be elaborated in the following subsections.

2.1. The APQ module

The APQ module customizes queries for each sample, improving segmentation performance on diverse cross-center medical data. The module obtains sample-related queries based on the image feature map F_e and the original semantic query Q_{ori}. Specifically, our APQ module uses a pixel classifier to predict a coarse mask from the encoder features F_e. This mask then filters F_e, and the result is globally averaged to produce the prompt feature F_p, as shown in Eq. 1.

$$F_p = GAP\left(sigmoid(\Phi(F_e)) \odot F_e\right), \tag{1}$$

where GAP stands for global average pooling, Φ denotes a pixel-level classifier, and $\odot$ represents the dot product.

The cross-attention output is merged with the original semantic query Q_{ori} to obtain the prompt semantic query Q_seg as in Eq. 2 and illustrated in Fig.2.

$$Q_{seg} = \text{Cross_Attn}(Q = Q_{ori}, K = F_p, V = F_p). \tag{2}$$

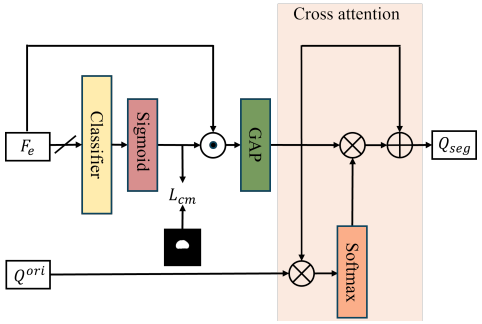


Fig. 2: An illustration of the adaptive prompt query module. $\nrightarrow$ indicates that the gradient does not propagate back.

2.2. The MQETD module

Boundary-oriented query: In cross-center segmentation, false negatives often appear at target boundaries, causing blurry edges. To address this, we introduce a boundary-oriented query Q_{bo} for precise boundary prediction, supervised by a boundary mask Y_{bo}. Specifically, the boundary mask is directly derived from the corresponding semantic label Y_{seg} following [21]. Morphological dilation and erosion operations are applied to Y, and the boundary region is obtained by subtracting the eroded mask from the dilated mask. Finally, a Gaussian filter is applied to smooth the boundary representation. The boundary mask is computed as:

$$Y_{bo} = G\big(\text{Dilate}(Y_{seg}) - \text{Erode}(Y_{seg})\big), \tag{3}$$

where $\text{Dilate}(\cdot)$ and $\text{Erode}(\cdot)$ represent the morphological dilation and erosion operations, respectively, and $G(\cdot)$ denotes a Gaussian filter on the binary mask.

True-false-positive query: We introduce a true-false-positive query (Q_{tfp}) to mitigate false positives that arise from ambiguous backgrounds in cross-center data. To effectively train this query, we employ a dynamic supervision

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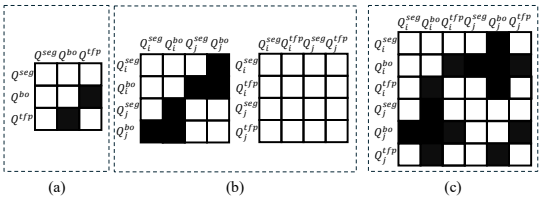


Fig. 3: Masked attention mechanism among Queries. (a) Masked-attention mechanism within semantics. (b) Masked-attention mechanism among semantics. (c) Unified masked-attention mechanism for multi-semantic interactions. i and j represent different semantics. Black squares indicate positions that are masked during the attention calculation process.

strategy. In each training iteration, the target label Y_{tfp} is generated by taking the union of the current semantic prediction (M_{seg}) and the ground truth label Y_{seg}(Eq. 4). This approach compels the Q_{tfp} to focus on both the true foreground and potential false-positive regions, thereby enhancing the model's ability to differentiate between them.

$$Y_{tfp} = \left\{ \begin{array}{ll} 1 & \text{if sigmoid}(M_{seg}) \geq 0.5 \\ 0 & \text{otherwise} \end{array} \right\} + Y_{seg}. \quad (4)$$

Notably, the final masks M_{seg} during training and inference are composed of masks obtained from the semantic query $Q^{'}_{ori}$ and the boundary-oriented query Q_{bo}, without filtering the masks predicted by the query Q_{tfp}.

Masked attention mechanism: To account for interrelationships in the multi-query strategy, we employ masked-attention mechanisms both within and across semantic categories to stabilize and optimize each query. Within the same semantic category, queries with different prediction targets should not freely interact in self-attention as illustrated in Fig. 3(a). Specifically, since there is no correlation between Q_{bo} and Q_{tfp}, we mask the attention calculation between these two queries to prevent unnecessary interaction. Cross-category attention is constrained to handle multi-semantic associations. As shown in Fig. 3(b), we mask attention between the boundary-oriented query Q_{bo} and the prompt semantic query Q_{seg} of other categories, since their boundaries are unrelated. In contrast, because the true-false-positive-related query Q_{tfp} may encode other-category or background signals, it is allowed to attend to of Q_{seg} all categories without masking. As shown in Fig. 3(c), the principles illustrated in Fig. 3(a) and Fig. 3(b) can be seamlessly extended to multi-semantic scenarios.

3. EXPERIMENTS AND ANALYSIS

3.1. Implementation Details

We conducted all experiments in PyTorch on a single NVIDIA GeForce RTX 3090 GPU. We employ ResNet-34 as the encoder to build the segmentation network with a U-shaped architecture. All other data processing protocols follow those

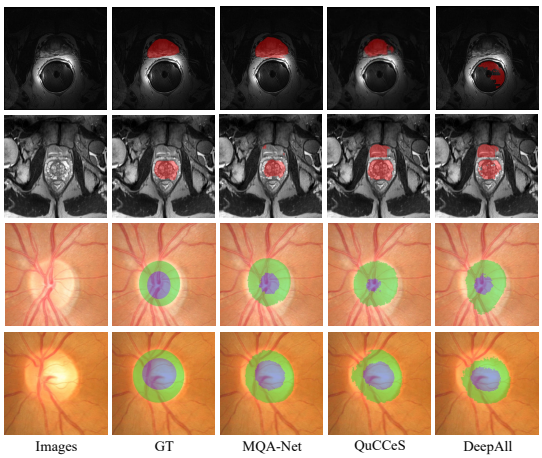


Fig. 4: Qualitative results of MQA-Net (ours) and DeepAll on the two segmentation tasks, together with ground truth.

used in TriD [7]. We utilize a combination of cross-entropy and Dice loss as the loss function for segmentation loss L_{seg}, coarse mask loss L_{cm}, boundary loss L_{bo}, and true-false-positive loss L_{tfp} denoted as Eq. 5.

$$L_{total} = L_{seg} + L_{cm} + 0.5 \times L_{bo} + 0.5 \times L_{tfp}. \quad (5)$$

3.2. Datasets and evaluation metrics

We validate the proposed method on two widely used multi-domain generalization datasets (Prostate and OD/OC).

Prostate: A dataset of 116 T2-MRI cases from six centers [22]. Models are trained on 2D slices (384×384) and evaluated in 3D using the Dice Similarity Coefficient (DSC) and Average Surface Distance (ASD).

OD/OC: A collection of fundus images from five public datasets [30–33] for optic disc (OD) and optic cup (OC) segmentation, evaluated with DSC on 512×512 images.

3.3. Comparative results in Prostate segmentation

Table 1 presents the prostate segmentation performance of MQA-Net, comparing it with other methods. Our method consistently delivers stable improvements, showing excellent generalization in different centers, resulting in improvements of 2.66% compared to DeepAll. This is particularly evident in domains with significant inter-domain differences, such as Domain 3 and Domain 5. This can be attributed to MQA-Net's focus on addressing the specific challenges of generalization in medical image segmentation, which not only enhances its generalization but also mitigates common issues such as blurry boundaries and background heterogeneity.

Fig. 4 qualitatively displays the performance of the model on the prostate dataset. The proposed MQA-Net provides clearer boundaries, significantly reduces false positives, and demonstrates a more precise ability to distinguish between foreground and background.

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Table 1: Quantitative results on prostate segmentation. The best results are highlighted with **bold**. The second best result is emphasized with a dashed line.

Models	Domain 1		Domain 2		Domain 3		Domain 4		Domain 5		Domain 6		Average	
	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓	DSC↑	ASD↓
DeepAll	91.32	0.81	89.09	0.73	83.22	3.46	88.99	0.70	80.21	6.93	89.52	1.00	87.08	2.27
BigAug [11]	90.68	1.80	89.52	1.00	84.86	**1.86**	89.04	1.59	73.24	5.94	89.1	1.16	86.07	2.23
SAML [22]	91.00	1.26	89.26	1.12	85.76	1.87	89.60	1.21	81.6	3.29	89.91	0.96	87.86	1.62
FedDG [6]	91.41	1.29	89.95	0.97	85.10	2.63	89.13	1.51	76.69	4.52	90.63	1.03	87.15	1.99
DoFE [23]	89.79	1.33	87.42	1.57	84.90	2.13	88.56	1.52	86.47	1.93	87.72	1.33	87.48	1.64
DCAC [24]	90.51	0.98	88.18	1.34	84.35	4.05	88.32	0.83	87.01	2.73	89.95	0.64	88.05	1.76
RandConv [25]	90.21	1.01	88.59	0.76	84.18	3.39	88.40	0.73	86.80	2.58	89.17	0.76	87.89	1.54
DSU [26]	90.92	0.79	88.19	0.82	84.57	3.86	88.68	0.73	86.25	1.80	89.13	0.70	87.96	1.45
EFDM [27]	91.39	0.81	90.13	0.69	85.13	4.02	89.81	0.66	84.52	4.59	90.08	0.70	88.51	1.91
MixStyle [28]	91.58	0.69	89.79	0.72	83.84	3.63	89.69	0.66	84.94	5.55	90.15	0.73	88.33	2.00
TriD [7]	91.61	0.68	89.97	0.68	85.10	3.10	89.26	0.73	87.15	6.26	89.79	0.74	88.81	2.03
QuCCeS [8]	91.72	0.71	90.29	0.67	86.48	3.21	90.02	0.66	86.95	**1.40**	90.64	0.63	89.35	1.21
SAMed [29]	89.94	1.16	90.32	1.15	80.13	4.00	87.88	1.28	85.25	3.50	**91.21**	0.63	87.46	1.95
MQA-Net (Ours)	91.88	**0.65**	**90.51**	**0.66**	**87.10**	2.99	**90.24**	**0.65**	**87.81**	1.47	90.91	**0.58**	**89.74**	**1.17**

Table 2: Quantitative results on OD/OC segmentation.

Models	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5	Average
	DSC↑	DSC↑	DSC↑	DSC↑	DSC↑	DSC↑
DeepAll	94.42/78.41	93.02/76.84	91.22/80.85	84.95/75.96	93.44/82.86	85.20
SAN-SAW [34]	93.34/76.31	92.88/**82.65**	90.78/81.18	88.07/77.61	93.43/83.97	86.02
DCAC [24]	94.43/76.72	93.70/79.21	91.05/81.23	88.12/77.87	95.71/85.32	86.33
RandConv [25]	93.11/76.21	92.50/81.33	89.01/81.33	88.33/76.56	95.32/85.16	85.89
DSU [26]	93.71/77.48	91.79/81.65	92.11/79.78	87.96/76.76	93.58/84.91	85.97
EFDM [27]	93.81/77.37	92.79/80.15	91.31/80.92	89.45/78.08	93.32/83.80	86.10
MixStyle [28]	94.07/79.85	93.33/80.32	91.73/80.94	89.44/78.00	93.71/82.61	86.40
TriD [7]	94.28/79.32	93.60/82.50	92.49/81.47	89.19/78.76	95.03/85.42	87.21
QuCCeS [8]	94.23/80.01	93.91/81.92	93.35/82.27	**93.47/83.18**	94.72/83.79	88.09
SAMed	94.60/76.20	92.10/82.32	92.79/**82.30**	72.03/50.92	**95.79/86.65**	82.57
MQA-Net (Ours)	**95.16**/80.29	**94.40**/82.50	**93.84**/81.40	91.98/83.13	95.57/83.85	**88.21**

Table 3: Performance comparison for each module of MQA-Net on the OD/OC segmentation.

MQETD	APQ	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5	Average
✗	✗	94.45/79.33	94.09/79.84	91.82/77.52	90.57/79.45	94.84/83.35	86.53
✗	✓	94.49/78.98	94.37/80.24	92.53/80.04	**93.05**/82.58	95.00/83.74	87.50
✓	✗	94.98/80.24	**94.73**/81.13	93.34/80.83	92.07/82.36	95.11/**84.28**	87.91
✓	✓	**95.16/80.29**	94.40/**82.50**	**93.84/81.40**	91.98/**83.13**	**95.57**/83.85	**88.21**

3.4. Comparative results in OD/OC segmentation

Table 2 presents OD/OC segmentation performance of MQA-Net and other state-of-the-art methods. MQA-Net improves generalization performance by 3.01% over the DeepAll and surpasses many SOTA approaches owing to its query-guided segmentation mechanism and targeted solutions for cross-center medical image segmentation. Fig. 4 qualitatively show joint segmentation results of the OD/OC dataset. The figure illustrates the superior performance of MQA-Net in handling unclear boundaries and false positive issues.

3.5. Ablation studies

We conducted comprehensive ablation studies to validate the effectiveness of our proposed Adaptive Prompt Query (APQ) and Multi-Query Enhanced Transformer Decoder (MQETD) modules. As shown in Table 3, both modules individually improve generalization compared to a standard query-guided decoder, with their combination yielding the best performance.

Table 4: Performance analysis for each component of the MQETD module on the OD/OC segmentation.

Q_{bo}	Q_{tfp}	Masked Attn.	Average
✗	✗	✗	86.53
✓	✗	✗	87.32
✓	✗	✓	87.64
✗	✓	✗	87.74
✗	✓	✓	87.52
✓	✓	✗	87.73
✓	✓	✓	**87.91**

A detailed component analysis of the MQETD on the OD/OC dataset (Table 4) further demonstrates its efficacy. Starting from a baseline of 86.53% DSC score, adding the boundary-oriented query(Q_{bo}) or the true-false-positive query (Q_{tfp}) individually boosted performance. Ultimately, integrating both specialized queries with the correctly designed masked attention achieved the top performance.

4. CONCLUSION

This paper addresses key challenges in cross-center medical image segmentation, with specific issues manifesting as evident domain shifts across centers, leading to increased background heterogeneity that causes false positives, and ambiguous target boundaries resulting in false negatives. To tackle these issues, we investigate the potential of adaptive and multiple queries to enhance segmentation performance. Generally, we introduce the Multi-Query Adaptive Network (MQA-Net), which consists of an Adaptive Prompt Query (APQ) module and a Multi-Query Enhanced Transformer Decoder (MQETD) module. Extensive experiments demonstrate that MQA-Net exhibits excellent cross-center generalization capability. Future work will focus on integrating multi-modal information to tackle more challenging multi-semantic, cross-center 3D medical image segmentation tasks.

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